

Section 09 — Triggers

BEFORE INSERT Trigger

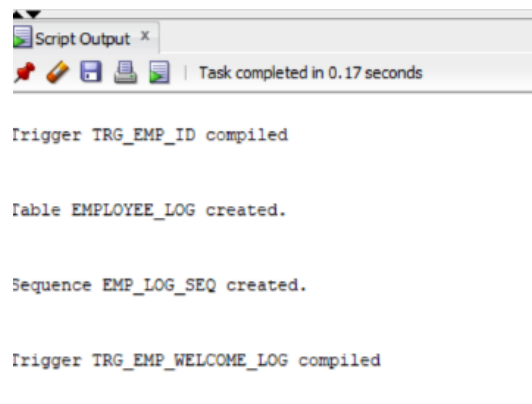
```
CREATE OR REPLACE TRIGGER TRG_EMP_ID
BEFORE INSERT ON EMPLOYEE
FOR EACH ROW
BEGIN
    IF :NEW.emp_ID IS NULL THEN
        :NEW.emp_ID := EMP_SEQ.NEXTVAL;
    END IF;
END;
/
```

AFTER INSERT Trigger

```
CREATE TABLE EMPLOYEE_LOG (
    log_id          NUMBER PRIMARY KEY,
    emp_id          NUMBER,
    action          VARCHAR2(50),
    log_timestamp   DATE
);

CREATE SEQUENCE EMP_LOG_SEQ START WITH 1 INCREMENT BY 1;
```

```
CREATE OR REPLACE TRIGGER TRG_EMP_WELCOME_LOG
AFTER INSERT ON EMPLOYEE
FOR EACH ROW
BEGIN
    INSERT INTO EMPLOYEE_LOG
    (log_id, emp_id, action, log_timestamp)
    VALUES
    (EMP_LOG_SEQ.NEXTVAL, :NEW.emp_ID, 'NEW HIRE', SYSDATE);
END;
/
```



Test — Insert Without emp_ID

```
INSERT INTO EMPLOYEE
(fname, lname, gender, age, contact_add, emp_email, emp_pass, job_ID, salary_ID)
VALUES ('Sara', 'Alsaadi', 'F', 28, 'Sohar', 'sara.trigger@ems.com', 'pass123', 5, 5);
```

```
INSERT INTO EMPLOYEE
(fname, lname, gender, age, contact_add, emp_email, emp_pass, job_ID, salary_ID)
VALUES ('Nasser', 'Alharthi', 'M', 32, 'Muscat', 'nasser.trigger@ems.com', 'pass123',
1, 1);
```

```
COMMIT;
```

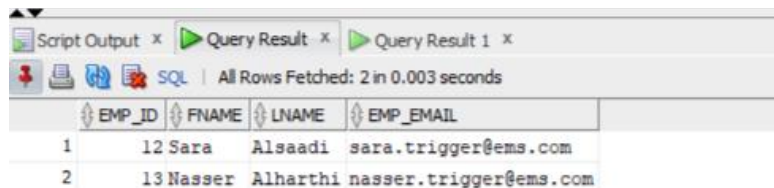
```
1 row inserted.
```

```
1 row inserted.
```

```
Commit complete.
```

Test — Verify emp_ID Was Auto-Generated

```
SELECT emp_ID, fname, lname, emp_email
FROM EMPLOYEE
WHERE emp_email IN ('sara.trigger@ems.com', 'nasser.trigger@ems.com');
```

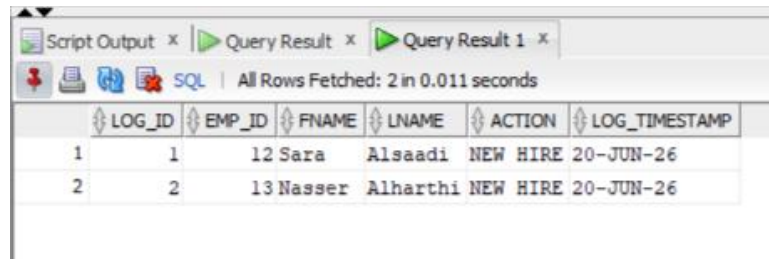


The screenshot shows a database query result window with the following data:

	EMP_ID	FNAME	LNAME	EMP_EMAIL
1	12	Sara	Alsaadi	sara.trigger@ems.com
2	13	Nasser	Alharthi	nasser.trigger@ems.com

Test — Verify NEW HIRE Log Was Created

```
SELECT l.log_id, l.emp_id, e.fname, e.lname,  
       l.action, l.log_timestamp  
FROM EMPLOYEE_LOG l  
JOIN EMPLOYEE e ON l.emp_id = e.emp_ID  
WHERE e.emp_email IN ('sara.trigger@ems.com', 'nasser.trigger@ems.com')  
ORDER BY l.log_id;
```



The screenshot shows a SQL query result window with the following data:

LOG_ID	EMP_ID	FNAME	LNAME	ACTION	LOG_TIMESTAMP
1	12	Sara	Alsaadi	NEW HIRE	20-JUN-26
2	13	Nasser	Alharthi	NEW HIRE	20-JUN-26